Semester 2 Algebra 2 Finals Review

Pre-Reqs (things you should be able to do!)

1. Factor completely.
   a. $2x^2 - 15x + 7$ b. $4x^3 - 16x^2$ c. $4x^2 + 10x - 24$ d. $3x^2 - x - 10$ e. $x^2 - 12x + 35$

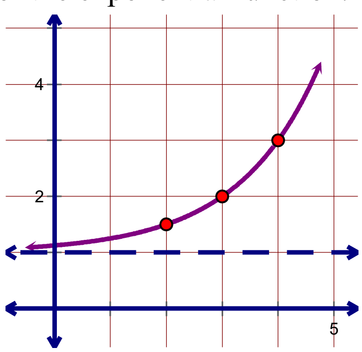
2. Simplify.
   a. $-2(x-1)(x+4)(x-5)$ b. $(x+7)(x-9)$ c. $-4(x-1)(x+7)$

3. Solve by factoring.
   a. $3x^2 - x - 10 = 0$ b. $x^2 - 16 = 0$ c. $3x^3 - 16x^2 + 5x = 0$ d. $x^2 - 12x + 35 = 0$

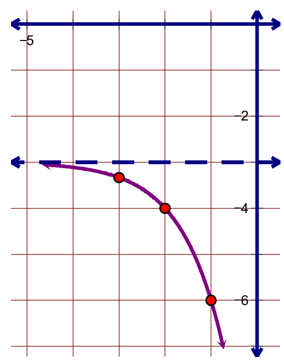
Units 6-7: Exponents & Logs

Write the equation for the exponential function.

31. Base 2



32. Base 3



Describe the transformations of the parent function. Then graph the transformed function and state the domain, range and end behavior.

33. $a(x) = \left(\frac{1}{2}\right)^x + 8$ 34. $b(x) = -2^{x-1} + 6$ 35. $c(x) = 3^{\frac{x}{2}} - 1$

Solve for x. Find the exact and approximate answer when appropriate.

36. $\sqrt[5]{x^3} = 27$ 37. $25^x = 125^{2x-5}$ 38. $2^x = \frac{1}{64}$ 39. $x^{\frac{3}{2}} = 8$

40. $9^x = 13$ 41. $8^{x+5} = 42$ 42. $3(7)^{x-4} = 240$ 43. $\ln(x+7) = 5$

44. $\log_7 343$ 45. $\log(x^2 - 10) = \log(3x)$ 46. $\ln(3x-7) = \ln(4x-10)$

47. The value of a new truck is $\$32,000$ and loses 8.4% of its value each year. How much will the truck be worth in 3 years?

48. An investment service promises to quadruple your money in 10 years. If you have $\$1000$ to invest, what is the annual interest rate (assuming annual compounding – not continuous)?

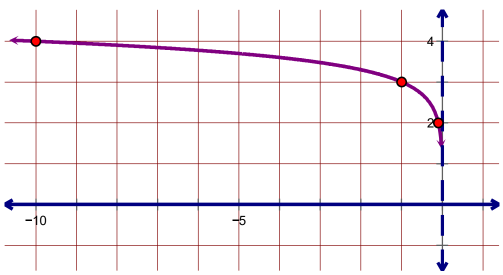
49. Sally opens a bank account by depositing $\$1250$. If her account will earn 4.75% interest, compounded quarterly, what will her balance be after 8 years?

50. Joe currently pays a $\$425$ premium for health insurance. If the premium increases at an annual rate of 7.2% each year, how many years will it take for the premium to be $\$1205.90$?

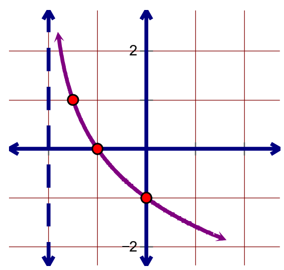
51. You have $\$2000$ to invest in an account that earns 7.5% interest, compounded continuously. Write an exponential equation to model the situation, then find when the account will be worth $\$7500$.

Write the equation for the logarithmic graph.

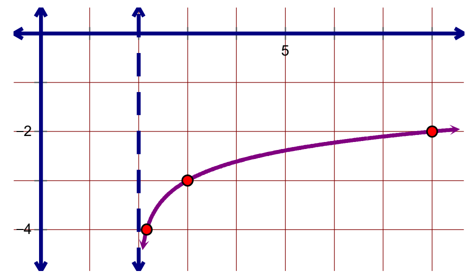
52. Base 10



53. Base 2



54. Base 6



Change to exponential or logarithmic form.

55. $\log_w y = z$ 56. $a^b = c$

Evaluate.

57. $\log 1000$ 58. $\log_2\left(\frac{1}{8}\right)$ 59. $\log 1$ 60. $\log_4 16$

Graph the following equations and state the domain, range and end behavior.

61. $a(x) = \log_3(x-4) - 2$ 62. $b(x) = -\log_2(x+3)$ 63. $c(x) = \log(-x) + 4$

64. Write the explicit rule for each of the following:

a.) $18, 6, 2, \frac{2}{3}, \ldots$ b.) $3, 6, 12, \ldots$ c.) $3072, 768, 192, \ldots$

65. Write the following using sigma notation:

a.) $5 + 17.5 + 61.25 + \ldots$ b.) $16 + 11.2 + 7.84 + \ldots + a_{13}$

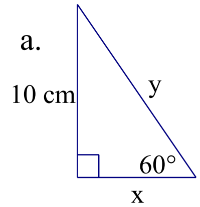
66. Find the sum of the following:

a.) $18, 6, 2, \frac{2}{3}, \ldots, a_7$ b.) $16 + 11.2 + 7.84 + \ldots + a_{13}$ c.) $\sum_{n=1}^{9} 2.1(3)^{n-1}$ d.) $\sum_{n=1}^{12} -3(1.5)^{n-1}$

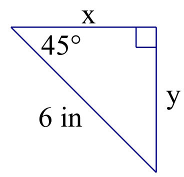
Units 7-8: Trig

67. A right triangle ΔANT has $A = 47^\circ$ and hypotenuse $n = 42$ ft. Solve the triangle.

68. Solve for x and y. Leave your answers in exact form. a.



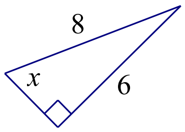
b.



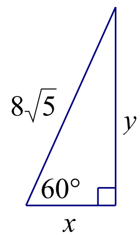
69. Simplify.

a. $\dfrac{12}{\sqrt{5}}$ b. $\dfrac{5\sqrt{3}}{\sqrt{15}}$

70. Given the triangle find x:



71. Find x and y.



72. Convert to radians or degrees.

a. $\dfrac{5\pi}{9}$ b. 72°

73. Sketch the angle on a set of axes, then find the reference angle.

a. $\dfrac{10\pi}{3}$ b. $-\dfrac{11\pi}{6}$ c. 137°

Find one positive and negative coterminal angle for the following angles.

74. $\theta = \dfrac{9\pi}{10}$ 75. $\theta = 283^\circ$ 76. $\theta = -\dfrac{\pi}{7}$

77. Find the exact value.

a. $\sin\dfrac{3\pi}{4}$ b. $\tan 300^\circ$ c. $\sin\dfrac{7\pi}{6}$ d. $\cos\dfrac{3\pi}{2}$ e. $\tan\left(-210^\circ\right)$ f. $\sin\dfrac{23\pi}{6}$

g. $\csc 210°$ h. $\sec\left(-\dfrac{5\pi}{4}\right)$ i. $\cot(-300°)$ j. $\sec 330°$ k. $\csc\dfrac{17\pi}{4}$ l. $\cot 1620°$

78. Given the range $[0, 2\pi]$ or $\left[0^\circ, 360^\circ\right]$, find <u>all</u> values of θ that make the statement true.

a. $\cos\theta = -\dfrac{\sqrt{2}}{2}$ (D) b. $\sin\theta = -\dfrac{\sqrt{3}}{2}$ (R) c. $\cos\theta = 0$ (R)

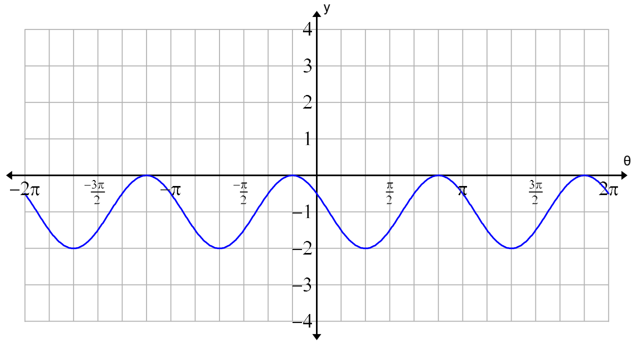
d. $\sin\theta = -\dfrac{1}{2}$ (D) e. $\cos\theta = \dfrac{\sqrt{3}}{2}$ (R) f. $\sin\theta = -1$ (D)

g. $\csc\theta = 2$ (D) h. $\cot\theta = -\sqrt{3}$ (R) i. $\sec\theta = -\sqrt{2}$ (R)

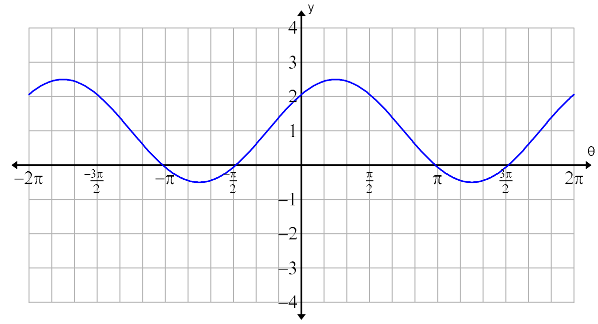
j. $\tan\theta$ is undefn. (D) k. $\csc\theta$ is undefn. (D) l. $\sec\theta = \dfrac{2\sqrt{3}}{3}$ (R)

79. Describe the transformations and find the equation of the graph using the indicated function.

a. $y = \cos\theta$



b. $y = \sin\theta$



80. List the amplitude, axis, period, and phase shift, then sketch a complete graph of each function.

a. $f(x) = 2\sin\left(x - \frac{2\pi}{3}\right) + 1$ b. $g(x) = \cos\left(\frac{x}{4}\right) - 2$ c. $h(x) = 3\cos\left(\frac{\pi}{3}x\right)$

81. Burntcoat Head in Nova Scotia, Canada, is known for its extreme fluctuations of tides. One day in April, the first high tide rose to 13.25 feet at 4:30 am. The first low tide at 1.85 feet occurred at 10:15 am. The second high tide was recorded at 4pm.

a. Sketch a graph.
b. Write an equation modeling the tides in terms of *x*, the number of hours since midnight.
c. What is the vertical shift and what does it represent?
d. What is the amplitude and what does it represent?

82. Naturalists find that the populations of some kinds of predatory animals vary periodically. Assume that the population of foxes in a certain forest varies sinusoidally with time. Records started being kept when time $t = 0$ years. A minimum number of foxes, 200 foxes, existed at $t = 2.9$ years. The next maximum, 800 foxes, occurred at $t = 5.1$ years.

a. Sketch a graph.

b. Write an equation expressing the number of foxes as a function of time.

c. Predict the population when $t = 6.2$.

83. Given $\cot\theta = \frac{7}{4}$ find the other 5 trig functions.

84. Given $\cos\theta = -\frac{5}{24}$ and $\sin\theta < 0$ find $\tan\theta$

85. Given $\sin\theta = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the other five trig functions.

86. Given $\csc\theta = \frac{3}{2}$ and θ is in QII, find $\sec\theta$.

Unit 10: Conics

87. Graph the following and identify the key features:

a.) $8x^2+8y^2=72$

b.) $(x+3)^2+(y-1)^2=49$

c.) $x^2+36y^2=36$

d.) $\frac{(x+5)^2}{25}+\frac{(y-1)^2}{81}=1$

e.) $16x^2-9y^2=144$

f.) $\frac{(y-3)^2}{4}-\frac{(x-4)^2}{36}=1$

g.) $(x+4)^2=-8(y-5)$

h.) $\frac{1}{20}y^2=x+1$

88. Write the equation of the conic in standard form with the information given.

a.) center $(3,4)$ radius = 5	b.) center $(-2,7)$ radius = $5\sqrt{3}$	c.) center $(-4,-1)$ endpoint $(2,3)$
d.) endpoints of diameter $(16,-2)$ and $(2,-8)$	e.) ellipse with center $(1,-6)$ a = 8, b = 4	f.) ellipse with vertices $(-3,12),(-3,-10)$ and covertices $(4,1),(-10,1)$
g.) ellipse with vertices $(8,9),(-2,9)$ and foci $(7,9),(-1,9)$	h.) ellipse with covertices $(4,-3),(-8,-3)$ and foci $(-2,5),(-2,-11)$	i.) hyperbola with center $(-2,-5)$ and a = 2, b = 3
j.)_hyperbola with vertices $(9,0),(-7,0)$ and covertices $(1,6),(1,-6)$	k.) hyperbola with vertices $(3,-5),(3,-13)$ and foci $(3,-4),(3,-14)$	l.) hyperbola with covertices $(7,16),(7,-8)$ and foci $(20,4),(-6,4)$
m.) parabola with vertex $(7,-5)$ and focus $(6,-5)$	n.) parabola with vertex $(-2,2)$ and directrix y = 5	o.) parabola with focus $\left(-\frac{11}{2},-1\right)$ and directrix passing through $\left(\frac{3}{2},-1\right)$

89. Write the equation in standard form ***and*** identify the conic section:

a.) $9x^2+y^2-54x-2y+46=0$

b.) $x^2+y^2-8x-4y-5=0$

c.) $x^2-10x+4y+33=0$

d.) $44x^2-y^2+24x-16y-32=0$

Unit 11: Statistics

See/revisit the Unit 11 Test review that we JUST completed! 😊